PPP1R13L-Related Arrhythmogenic Cardiomyopathy: A Comprehensive Disease Characterization

Disease: PPP1R13L-Related Arrhythmogenic Cardiomyopathy (PPP1R13L cardiocutaneous syndrome) **Causal gene: PPP1R13L** (protein phosphatase 1 regulatory subunit 13 like; encodes **iASPP**, the inhibitor of apoptosis-stimulating protein of p53; also known as RAI, NKIP1) **Key identifiers:** OMIM:620519 · MONDO:0957795 · HGNC gene PPP1R13L · Chromosome 19q13.2–q13.32 **Category:** Mendelian, autosomal-recessive

Summary

PPP1R13L-related arrhythmogenic cardiomyopathy is an **ultra-rare, autosomal-recessive cardiocutaneous syndrome** caused by **biallelic loss-of-function variants in PPP1R13L**, the gene encoding **iASPP** (inhibitor of apoptosis-stimulating protein of p53, also historically called NKIP1/RAI). The disease is distinct from the many serially numbered cardiomyopathies: the causal gene here is unambiguously **PPP1R13L**, not a desmosomal plakin or a sarcomeric gene, although the downstream lesion converges on the cardiac desmosome. To date only approximately **15–20 affected children** have been reported worldwide, typically from consanguineous or founder pedigrees. The cardinal manifestation is **severe, early-onset, rapidly progressive dilated/arrhythmogenic cardiomyopathy** (mean onset ~2 years 10 months; range 3 months–9 years) with ventricular arrhythmias and high early-childhood mortality, accompanied by a **variably expressed ectodermal phenotype** (woolly/sparse hair, abnormal/pointed teeth, skin abnormalities) and, in some families, orofacial clefting and ocular disease (glaucoma, keratitis).

The **central disease mechanism** is loss of iASPP from the cardiomyocyte **intercalated disc**, where iASPP normally binds **desmoplakin and desmin** and maintains the integrity of the **desmosome–intermediate-filament network**. Its loss destabilizes desmin anchoring at the intercalated disc, producing an arrhythmogenic right-ventricular cardiomyopathy (ARVC)-like phenotype with sudden cardiac death — demonstrated concordantly in **human explanted hearts**, **Ppp1r13l-mutant mice**, and naturally occurring **Poll Hereford cattle** cardiomyopathy

and woolly-haircoat syndrome. Two additional mechanistic arms amplify disease: dysregulated **NF-κB / IL-1 inflammation** (iASPP is a common inhibitor of RelA/p65 and p53) and loss of **Keap1–Nrf2 antioxidant protection**, which sensitizes cardiomyocytes to oxidative/hypoxic injury. The ectodermal features are attributable to iASPP's regulation of **p63**, which is required for epithelial stratification.

Diagnosis rests on exome or genome sequencing (biallelic-variant filtering) combined with ACM/DCM cardiac workup (echocardiography, cardiac MRI with late gadolinium enhancement, ECG/Holter, and explant/autopsy histopathology). **Management** is supportive: heart-failure pharmacotherapy, antiarrhythmics, implantable cardioverter-defibrillator (ICD), and definitive **heart transplantation** for end-stage disease. **Investigational IL-1 blockade** reduces ventricular ectopy and normalizes serum cytokines but has not restored contractile function. Prognosis is poor without transplantation, with multiple deaths before age three in reported cohorts.

Section 1 — Disease Information

Overview. PPP1R13L-related arrhythmogenic cardiomyopathy (also termed **PPP1R13L cardiocutaneous syndrome** or **PPP1R13L-associated cardiocutaneous syndrome**) is a Mendelian, autosomal-recessive disorder in which biallelic loss of function of PPP1R13L produces a syndromic combination of **life-threatening childhood cardiomyopathy** (dilated and/or arrhythmogenic) plus **ectodermal abnormalities** of hair, skin, and teeth. The cardiomyopathy may show features of inflammation on histology. First described in humans in 2017 as a lethal infantile dilated cardiomyopathy in Arab Christian infants, the condition has since been expanded through international gene-matching efforts and single case reports.

Key identifiers: | Resource | Identifier | |---|---| | OMIM | **620519** (phenotype); PPP1R13L gene 607463 | | MONDO | **MONDO:0957795** | | Gene (HGNC) | **PPP1R13L** | | Protein (UniProt) | iASPP (Q8WUF5) | | Cytogenetic locus | 19q13.2–q13.32 |

Synonyms / alternative names: PPP1R13L cardiocutaneous syndrome; PPP1R13L-associated cardiocutaneous syndrome; PPP1R13L-related cardiomyopathy; iASPP-related cardiomyopathy. The gene carries legacy aliases **NKIP1** (NF-κB interacting protein 1) and **RAI** (RelA-associated inhibitor).

Information source. Knowledge is derived almost exclusively from **individual-patient clinical reports and small family cohorts** (case reports, GeneMatcher-facilitated case series, and explant/autopsy studies), supplemented by **model-organism** (mouse, cattle) and **in vitro** mechanistic work. There are no aggregated registry- or EHR-scale datasets because of the disease's rarity.

Section 2 — Etiology

Primary cause — genetic. The disease is caused by **biallelic (homozygous or compound-heterozygous) loss-of-function variants in PPP1R13L**. There is no infectious, toxic, or environmental primary cause. The GeneMatcher-facilitated international study identified biallelic PPP1R13L variants in **seven children from five unrelated families** with severe dilated cardiomyopathy following exome or genome sequencing, establishing a robust gene–disease association: *"biallelic variants in PPP1R13L were identified in seven children with severe DCM from five unrelated families following exome or genome sequencing"* (PMID: 32666529). The ClinGen DCM gene-curation expert panel's 2024–2025 reassessment of 68 genes includes PPP1R13L among curated DCM genes (PMID: 42708185).

Genetic risk factors. The disease is monogenic and recessive; the sole genetic requirement is two defective PPP1R13L alleles. **Consanguinity** and **founder alleles** are major contributors (see Section 9). Reported disease variants are predominantly **nonsense and frameshift** (loss-of-function) alleles.

Environmental risk factors. None established as causal. As with other arrhythmogenic cardiomyopathies, **physical exertion** and **sympathetic/adrenergic stress** are plausible arrhythmia triggers, and inflammatory flares may contribute, but no PPP1R13L-specific environmental risk factor has been demonstrated. Sex does not appear to modify primary risk (recessive, both sexes affected).

Protective factors. No genetic or environmental protective factors have been identified. Heterozygous carriers are generally unaffected, consistent with recessive inheritance and the absence of a dominant-negative effect for most LoF alleles.

Gene–environment interactions. Not formally characterized. The mechanistic convergence on **inflammation (NF-κB/IL-1)** and **oxidative stress (Nrf2/Keap1)** suggests that inflammatory or ischemic/hypoxic stressors could plausibly worsen myocardial injury in genetically susceptible individuals, but this remains **inferred, not demonstrated**.

Section 3 — Phenotypes

The phenotype is **syndromic and variably expressed**, even within families. Approximately 15 patients were tabulated in one 2024 review, which summarized that *"PPP1R13L is associated with a recessive syndrome leading to cardiac anomalies with skin, teeth, and hair abnormalities. Fifteen patients have been described so far"* (PMID: 39579152).

Cardiac (core, high penetrance)

- **Severe early-onset dilated cardiomyopathy / arrhythmogenic cardiomyopathy** — ventricular dilatation and systolic dysfunction; the dominant and most lethal feature.
- **Ventricular arrhythmias / premature ventricular contractions / ventricular tachycardia** — arrhythmogenic substrate, risk of sudden cardiac death.
- **Heart failure** — progressive, leading to death or transplant.
- Suggested HPO: **HP:0001644** (Dilated cardiomyopathy), **HP:0011663** (Right ventricular cardiomyopathy / arrhythmogenic), **HP:0004308** (Ventricular arrhythmia), **HP:0001635** (Congestive heart failure), **HP:0001645** (Sudden cardiac death).

Ectodermal (variably expressed)

- **Woolly / sparse / dry hair**, **high anterior hairline** — HP:0002216 (Woolly hair), HP:0008070 (Sparse hair).
- **Abnormal / pointed teeth** — dental anomalies.
- **Skin abnormalities** — abnormal epidermal morphology.
- **Broad nasal bridge**, dysmorphic features.

Orofacial and ocular (subset of families)

- **Cleft lip and palate** — HP:0000202.
- **Glaucoma** — HP:0000501; **keratitis** / ocular surface disease (neonatal keratitis also seen in the bovine model).

Other / expanded phenotype

- **Anorectal anomaly** reported in an extended-phenotype case (PMID: 39579152).
- **Global developmental delay** in one child (PMID: 37698259).

Characteristics. Age of onset: **neonatal to childhood** (cardiac mean ~2 y 10 m). Severity: **severe** for cardiac features; **mild–moderate and variable** for ectodermal features. Progression: cardiac disease is **progressive**; arrhythmias are **episodic**. Frequency: cardiomyopathy is near-universal in affected individuals; ectodermal features are present in most but **variably expressed**, so a given patient may have DCM without obvious skin/hair findings, which can obscure the syndromic diagnosis.

Intrafamilial variability is well illustrated in a consanguineous family where *"The index case had DCM and wooly hair, two of his siblings had DCM and CLP while three cousins had, in addition, glaucoma"* (PMID: 37698259).

Quality-of-life impact. Severe. Affected children face recurrent heart failure, arrhythmia, hospitalization, transplant evaluation, and frequently death in early childhood. No formal EQ-5D/SF-36/PROMIS data exist for this ultra-rare condition.

Section 4 — Genetic / Molecular Information

Causal gene. PPP1R13L (OMIM gene 607463), chromosome **19q13.2–q13.32**, 13 exons, encoding **iASPP**. iASPP contains ankyrin repeats and an SH3 domain mediating its protein–protein interactions with p53-family proteins and with desmosomal proteins (PMID: 18275817).

Pathogenic variants (representative). | Variant (protein / cDNA) | Type | Zygosity | Phenotype / note | PMID | |---|---|---|---|---| | Homozygous premature stop codon | Nonsense | Homozygous | 5 Arab Christian infants, 4 families; lethal infantile DCM (founder) | 28069640 | | p.Arg330ProfsTer76 | Frameshift | Homozygous | Consanguineous family, 6 affected children | 37698259 | | p.Gln194Ter (c.580C>T) | Nonsense | Homozygous | Arrhythmogenic cardiomyopathy | 35933355 | | Novel frameshift | Frameshift | Homozygous | 4-year-old, DCM + ectodermal features | 41919412 | | Multiple biallelic LoF (5 families) | Nonsense/ frameshift | Biallelic | GeneMatcher cohort, 7 children | 32666529 | | Bovine c.956_962dup7, p.Ser322GlnfsX4 | Frameshift | Homozygous | Poll Hereford cattle CWH model | 19016676 |

Variant classification. Reported disease alleles are overwhelmingly **pathogenic/likely pathogenic loss-of-function** (nonsense, frameshift) per ACMG/AMP criteria. The nonsense example was described as *"a novel homozygous stop-gain pathogenic variant, c.580C > T: p.Gln194Ter, in the PPP1R13L gene"* (PMID: 35933355). A heterozygous missense variant p.A620P was scored only as a **VUS** in an ACM screen where a co-occurring TP63 nonsense

variant was likely pathogenic, and the authors concluded PPP1R13L involvement in that case "remains to be determined" (PMID: 30453078) — underscoring that the established disease mechanism is **biallelic LoF**, not monoallelic missense.

Allele frequency. Causal alleles are extremely rare/private in population databases (gnomAD), consistent with an ultra-rare recessive disorder; some alleles are population-enriched founder variants.

Somatic vs germline. Germline exclusively.

Functional consequence. Loss of function (biallelic). Loss of iASPP protein removes its stabilizing role at the intercalated disc and its regulatory inhibition of p53/p63/NF-κB.

Modifier genes. None formally established. Given iASPP's binding partners, **desmoplakin, desmin, TP63, and NF-κB-pathway genes** are mechanistically plausible modifiers but unproven in humans.

Epigenetic information. No PPP1R13L-specific disease methylation/chromatin data available.

Chromosomal abnormalities. None; disease is caused by small sequence variants, not large structural rearrangements.

Section 5 — Environmental Information

No environmental, lifestyle, or infectious agent is a **cause** of this monogenic disease. Of note, PPP1R13L-related cardiomyopathy can **present as or mimic myocarditis** (inflammatory histology), which is relevant to differential diagnosis rather than etiology; some reports describe inflammation on cardiac histology (PMID: 35924320). Standard arrhythmia precautions (avoidance of extreme exertion) are advisable as for other arrhythmogenic cardiomyopathies, but this is extrapolated rather than PPP1R13L-specific.

Section 6 — Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic loss-of-function variant in PPP1R13L → leads to** absence/severe reduction of iASPP protein.

2. Loss of iASPP **at the cardiomyocyte intercalated disc → results in** failure to bind and stabilize **desmoplakin and desmin** (*"iASPP interacts with desmoplakin and desmin in cardiomyocytes to maintain the integrity of desmosomes and intermediate filament networks in vitro and in vivo"*, PMID: 25691752).
3. Destabilized desmosome–intermediate-filament network → **leads to loss of desmin anchoring at the intercalated disc** — the central lesion seen in human explants (*"loss of desmin-desmosome anchoring in cardiomyocytes appears to be the central mechanism of PPP1R13L cardiomyopathy"*, PMID: 41928611).
4. Weakened cell–cell adhesion under mechanical load → **results in** cardiomyocyte injury, an **ARVC-like structural phenotype** (RV dilatation; demonstrated at E16.5 in mice) and progressive ventricular dysfunction.
5. **Branch A (inflammation):** Loss of iASPP's inhibition of **RelA/p65 NF-κB → leads to** increased NF-κB signaling and **IL-1/cytokine-driven inflammation** (increased nuclear NF-κB in non-cardiomyocytes of human explants) → amplifies myocardial damage and arrhythmogenic remodeling.
6. **Branch B (oxidative stress):** Loss of iASPP → **results in** impaired **Keap1–Nrf2** antioxidant activation → cardiomyocytes become sensitized to hypoxia/reoxygenation injury, with **increased apoptosis and ROS** (*inferred* for human disease; demonstrated in rodent cardiomyocytes).
7. **Branch C (ectodermal):** Loss of iASPP's regulation of **p63 → leads to** defective epithelial stratification → **hair, skin, dental, and orofacial** abnormalities.
8. Convergence → **severe early-onset dilated/arrhythmogenic cardiomyopathy**, **ventricular arrhythmias**, **heart failure**, and **sudden cardiac death**.

Detail by category

Molecular pathways. Three interlocking axes: (1) **desmosomal / cytoskeletal** structural pathway (desmoplakin–desmin at the intercalated disc); (2) **NF-κB inflammatory** signaling — *"inhibitor of Apoptosis Stimulating p53 Protein (iASPP) is one such common regulator of both RelA/p65 and p53"* (PMID: 26646590); (3) **Keap1–Nrf2 (NFE2L2)** antioxidant signaling (PMID: 33332723). A fourth, **p53/p63** transcriptional regulation, underlies the ectodermal phenotype — *"iASPP is likely to control epithelial stratification by regulating p63's transcriptional activity"* (PMID: 21930934).

Cellular processes. Cell–cell adhesion/desmosome assembly, apoptosis regulation, inflammation, oxidative-stress response, and epithelial stratification/differentiation.

Protein dysfunction. Loss of function of iASPP — a shuttling multifunctional protein that in the nucleus inhibits p53/p63 and in the cytoplasm/intercalated disc regulates desmosome integrity. iASPP's ankyrin-repeat + SH3 architecture mediates differential binding to p53, p63, and p73 (PMID: 18275817).

Metabolic / oxidative changes. Impaired Nrf2-dependent antioxidant defense elevates ROS and apoptosis under hypoxia/reoxygenation: *"The upregulation of iASPP repressed H/R-induced injury by decreasing levels of apoptosis and reactive oxygen species production"* and *"iASPP ameliorates H/R-induced apoptosis and oxidative stress in cardiomyocytes via potentiating Nrf2 signaling via modulation of Keap1"* (PMID: 33332723).

Immune involvement. Dysregulated NF-κB → IL-1-driven inflammation; human explants show increased NF-κB nuclear signaling in non-cardiomyocytes (PMID: 41928611).

Tissue-damage mechanisms. Mechanical disruption of intercalated discs, fibrosis/fibrofatty remodeling (ACM-type substrate), inflammatory injury, and oxidative/apoptotic cardiomyocyte loss.

Molecular profiling. Human explant immunofluorescence demonstrates **loss of desmin at the intercalated disk** as the central feature distinguishing PPP1R13L hearts from non-PPP1R13L DCM controls, plus increased NF-κB in non-cardiomyocytes (PMID: 41928611).

Suggested ontology terms. GO: desmosome organization, cell–cell junction assembly, apoptotic process (**GO:0006915**), NF-κB signaling (**GO:0038061**), cellular response to oxidative stress (**GO:0034599**), regulation of transcription (**GO:0006355**). CL: cardiac muscle cell / cardiomyocyte (**CL:0000746**), keratinocyte (**CL:0000312**).

Section 7 — Anatomical Structures Affected

Organ level. Primary: **heart** (UBERON:0000948), especially the **ventricular myocardium** with right- and both-ventricular involvement. Secondary systems: **integument** (skin, hair follicles), **teeth / oral cavity**, **eye** (glaucoma, cornea). Body systems: **cardiovascular** (primary), **integumentary/ectodermal**, **orofacial/craniofacial**, **ocular/visual**.

Tissue and cell level. Cardiac muscle tissue and **cardiomyocytes** (CL:0000746), specifically at the **intercalated disc** where iASPP, desmoplakin, and desmin co-localize. Ectodermally derived **epithelial/keratinocyte** lineages (CL:0000312) account for hair/skin/tooth features.

Subcellular level. Intercalated disc / desmosome (GO:0030057 desmosome), **intermediate-filament cytoskeleton** (desmin network), **nucleus** (p53/p63/NF-κB transcriptional regulation), **cytoplasm** (iASPP shuttling).

Localization & lateralization. Cardiac involvement is **biventricular** with frequent **right-ventricular** structural change (ARVC-like). Ectodermal features are generally **bilateral/symmetric**.

Section 8 — Temporal Development

Onset. Congenital to early-childhood. Cardiac presentation is early: mean **2 years 10 months** (range **3 months–9 years**) (PMID: 32666529); the original cohort presented at **4–30 months** — *"Five Arab Christian infants, aged 4-30 months from four families, were diagnosed with DCM associated with mild skin, teeth, and hair abnormalities. All passed away before age 3"* (PMID: 28069640). Onset pattern is typically **subacute to chronic-progressive**, sometimes with an acute decompensated heart-failure presentation.

Progression. Rapidly progressive. DCM *"presented early (mean: 2 years 10 months; range: 3 months-9 years) and was progressive, resulting in death (n = 3) or transplant (n = 3), with one child currently awaiting transplant"* among 7 children (PMID: 32666529). Disease course is **progressive** for pump failure with **episodic** arrhythmic events.

Patterns. No spontaneous remission. The only "remission" is **treatment/transplant-induced** replacement of the diseased organ. The **critical intervention window** is early childhood — before irreversible pump failure — arguing for early genetic diagnosis, arrhythmia surveillance, and timely transplant evaluation.

Section 9 — Inheritance and Population

Inheritance. Autosomal recessive. Both biallelic homozygous (consanguineous/founder) and compound-heterozygous genotypes are reported.

Epidemiology. Ultra-rare; only ~15–20 patients reported worldwide. No formal prevalence/incidence estimates exist. The disease is likely **underdiagnosed** because its cardiac phenotype overlaps non-syndromic DCM and PPP1R13L is a recent addition to gene panels.

Penetrance / expressivity. Cardiomyopathy penetrance appears **high** in biallelic-LoF individuals; **ectodermal features are variably expressed** even among affected siblings/cousins (PMID: 37698259). Expressivity is therefore **variable**.

Founder effects / consanguinity. Strong role. The first human cohort — homozygous premature stop codon in **5 Arab Christian infants from 4 families** — is consistent with a **regional founder allele** (PMID: 28069640). A recurrent homozygous frameshift **p.Arg330ProfsTer76** segregated in a **consanguineous family with 6 affected children** (PMID: 37698259). Consanguinity is a recurring feature across pedigrees.

Genetic anticipation / germline mosaicism. No evidence (not a repeat-expansion disorder; recessive).

Carrier frequency. Not formally established; expected to be very low in general populations but potentially elevated in specific founder/consanguineous communities.

Demographics. Reported across Middle Eastern (Arab Christian), consanguineous, and other populations. **Both sexes affected** (recessive; no sex predilection). **Age distribution** skews to infancy/early childhood owing to high early mortality. The gene maps to chromosome **19q13.2–q13.32** (bovine orthologue on BTA18, PMID: 19016676).

Section 10 — Diagnostics

Genetic testing is the diagnostic cornerstone. All reported PPP1R13L cases were identified by **whole-exome or whole-genome sequencing** with biallelic-variant filtering (PMID: 32666529, PMID: 37698259). Single-gene testing rarely suffices because the presentation overlaps non-syndromic DCM and PPP1R13L is a newer panel addition; authors recommend that *"PPP1R13L should be included in ACM genetic testing to improve the identification of at-risk family members and the diagnostic yield"* (PMID: 35933355). WES/WGS with biallelic filtering and gene panels covering cardiomyopathy/ACM genes are the preferred approaches; CMA/karyotype/FISH/mitochondrial/repeat-expansion testing are **not applicable** (small sequence variants, nuclear recessive).

Cardiac workup (parallels ACM/DCM). - **Echocardiography** — ventricular dilatation and systolic dysfunction. - **Cardiac MRI with late gadolinium enhancement (LGE)** — the gold standard for ventricular structure/function, edema, and fibrosis: *"Cardiac magnetic resonance imaging (CMR) with late gadolinium enhancement (LGE) is the gold standard for evaluating left and right ventricular structure and function, edema, and fibrosis"* (PMID: 34970070); detects

fibrofatty/RV substrate. - **ECG / Holter monitoring** — ventricular arrhythmia burden, PVCs, VT. - **Histopathology** on explant/autopsy — **loss of desmin at the intercalated disk** and increased NF-κB in non-cardiomyocytes are characteristic (PMID: 41928611). - **Postmortem genetic testing** confirms concealed arrhythmogenic cardiomyopathy in sudden death, including PPP1R13L (PMID: 42098522).

Biomarkers. No specific circulating biomarker; serum **IL-1/IL-10** can be elevated and track with the inflammatory arm (and normalize with IL-1 blockade) (PMID: 41928611). Standard heart-failure biomarkers (BNP/NT-proBNP, troponin) apply generically.

Clinical criteria & differential diagnosis. Diagnosis integrates ACM/DCM imaging-electrophysiology criteria with **confirmatory biallelic PPP1R13L genotype**. Differentials: desmosomal ACM (DSP, PKP2, DSG2), other childhood DCM (sarcomeric, metabolic, mitochondrial), **myocarditis** (which PPP1R13L disease can mimic), and syndromic ectodermal disorders. The presence of **ectodermal features (woolly hair, dental/skin anomalies)** alongside early DCM should prompt PPP1R13L testing.

Screening. Cascade genetic testing of at-risk relatives and **carrier testing** in consanguineous families; prenatal/preimplantation testing is feasible when the familial variant is known.

Section 11 — Outcome / Prognosis

Poor, with high early-childhood mortality. In the 7-child cohort, outcomes were **3 deaths, 3 transplants, and 1 awaiting transplant** (PMID: 32666529). In the original cohort, all **5 infants died before age 3** (PMID: 28069640). In a consanguineous family with p.Arg330ProfsTer76, **all but one of 6 affected children died in early childhood** (PMID: 37698259).

Mortality mechanisms. Progressive **heart failure** and **sudden cardiac death** from ventricular arrhythmia (ARVC-like substrate).

Morbidity / QoL. Severe cardiac disability, recurrent hospitalization, transplant dependency; ectodermal and ocular comorbidity in subsets. No validated QoL instruments reported.

Recovery potential. No spontaneous recovery; **heart transplantation is definitive** for end-stage disease. IL-1 blockade reduced arrhythmia *"without contractile recovery"* (PMID: 41928611).

Prognostic factors. Early age at cardiac onset, rapid progression to heart failure, high ventricular-arrhythmia burden, and extent of LGE/fibrosis portend worse outcome (extrapolated from ACM literature; PMID: 39486037, PMID: 34970070).

Section 12 — Treatment

No disease-specific or curative pharmacotherapy exists. Management is supportive and mirrors ACM/DCM care.

Pharmacotherapy (supportive). Standard **heart-failure therapy** (as age-appropriate: ACE inhibitors/ARBs, beta-blockers, diuretics, aldosterone antagonists) and **antiarrhythmics** for ventricular arrhythmia control. NCIT concepts: heart-failure pharmacotherapy, antiarrhythmic agents.

Device / interventional. Implantable cardioverter-defibrillator (ICD) for sudden-death prevention given the arrhythmogenic substrate (NCIT: Implantable Cardioverter Defibrillator).

Definitive therapy. Heart transplantation — the definitive treatment for end-stage disease (PMID: 32666529). NCIT: Heart Transplantation.

Investigational / emerging — IL-1 blockade. Two pediatric patients (47 and 32 months) with refractory ventricular arrhythmias/heart failure received **anti-IL-1 immunomodulatory therapy**: *"IL-1 blockade was associated with reduced ventricular ectopy and normalized serum cytokine levels (IL-1, IL-10) but without contractile recovery"* (PMID: 41928611). This targets the NF-κB/IL-1 inflammatory arm and is the most promising mechanism-informed adjunct to date, though it does not reverse pump failure.

Pharmacogenomics / gene / cell / RNA therapy. None established. Gene-replacement is conceptually attractive (recessive LoF) but not developed.

Section 13 — Prevention

- **Primary prevention:** For a recessive Mendelian disease, prevention is reproductive/genetic — **genetic counseling, carrier screening in consanguineous/founder families, and prenatal or preimplantation genetic testing** when the familial variant is known.

- **Secondary prevention: Cascade genetic testing** of relatives, with early cardiac surveillance (echo/CMR/Holter) of genotype-positive children to enable timely heart-failure therapy, ICD, and transplant listing.
- **Tertiary prevention:** Arrhythmia management, ICD for sudden-death prevention, heart-failure optimization, and investigational IL-1 blockade to reduce arrhythmic burden.
- **Counseling:** Autosomal-recessive recurrence risk is **25%** per pregnancy for carrier couples; discuss consanguinity.
- **No immunization or public-health/environmental prevention applies** (monogenic disease).

Section 14 — Other Species / Natural Disease

PPP1R13L-related disease is notable for a **naturally occurring large-animal model**. **Poll Hereford cattle** develop **cardiomyopathy and woolly-haircoat syndrome (CWH)** caused by a PPP1R13L (NKIP1/RAI) frameshift — *"a 7-bp duplication (c.956_962dup7) in exon 6 of this 13-exon gene"*, p.Ser322GlnfsX4, on BTA18 (PMID: 19016676). The bovine phenotype strikingly parallels the human syndrome: **cardiomyopathy** + **woolly haircoat**, plus neonatal keratitis — making it a strong comparative-pathology model and one of the earliest clues linking PPP1R13L to a cardiocutaneous syndrome.

- **Taxonomy:** *Bos taurus* (NCBI Taxon 9913); *Mus musculus* (NCBI Taxon 10090).
- **Orthologous gene:** bovine/murine **Ppp1r13l**.
- **Comparative biology:** The cardiac desmosomal mechanism and the hair/skin phenotype are **evolutionarily conserved** across human, mouse, and cattle, supporting mechanistic generalizability.
- **Zoonotic potential:** None (genetic disease).

Section 15 — Model Organisms

Mouse. The **Ppp1r13l(Δ8/Δ8)** (exon-8 deletion) mouse — *"iASPP-deficient mice with exon 8 deletion (Ppp1r13l(Δ8/Δ8)) die of sudden cardiac death, displaying features of ARVC"* (PMID: 25691752). iASPP deficiency induces **right-ventricular dilatation at E16.5**, and iASPP interacts

with desmoplakin and desmin to maintain desmosome/intermediate-filament integrity *in vitro* and *in vivo*. The **Wa3** mouse is an additional allele. These models **recapitulate** arrhythmogenic cardiomyopathy and sudden death well, validating the desmosomal mechanism.

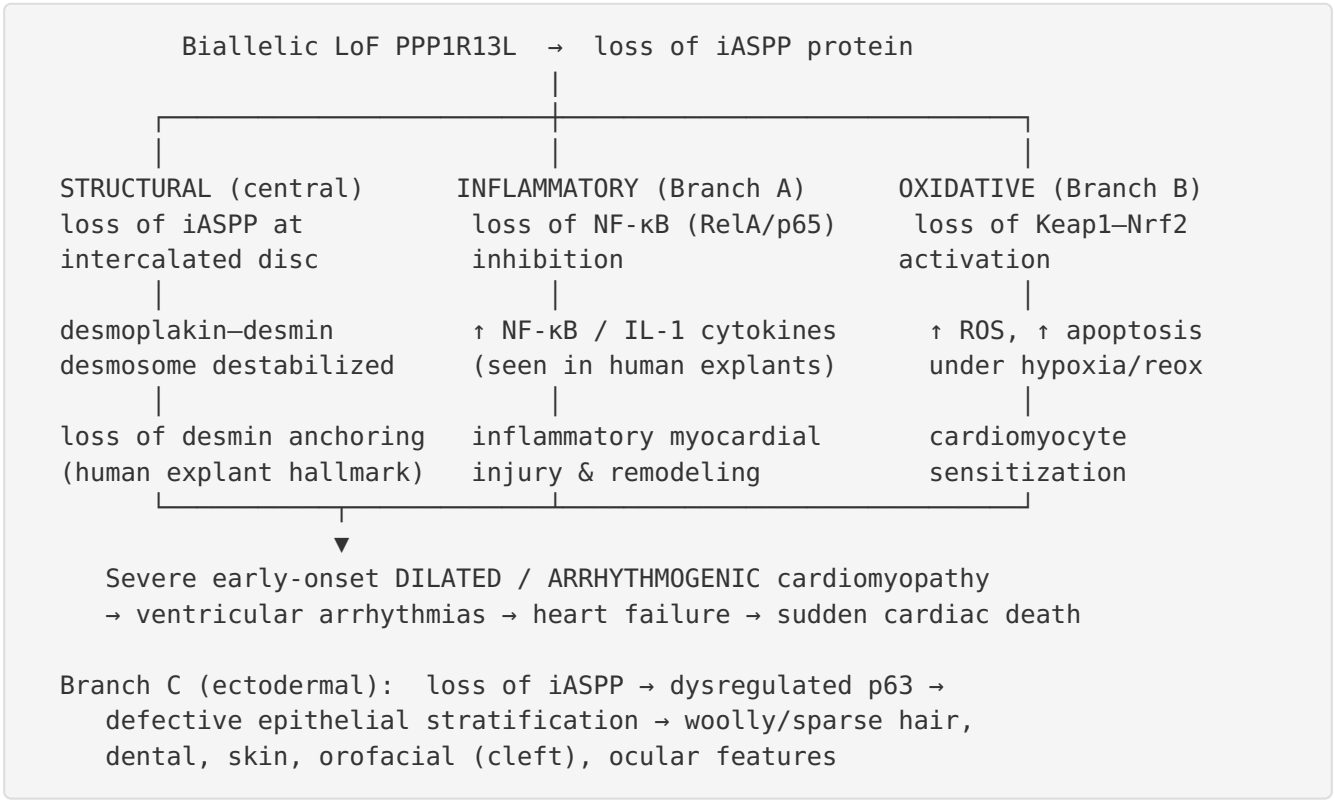
Cattle (natural model). Poll Hereford CWH (above) — excellent recapitulation of the combined **cardiac** + **cutaneous** phenotype (PMID: 19016676).

In vitro / cellular. Rodent cardiomyocyte hypoxia/reoxygenation systems demonstrate the **Nrf2/Keap1 cardioprotective** arm (PMID: 33332723); cancer-cell and epithelial systems established iASPP's **p53/p63/NF-κB** regulatory roles (PMID: 21930934, PMID: 26646590).

Model limitations. Mouse embryonic lethality/early death complicates longitudinal study; cellular systems address single pathways in isolation; no patient-derived iPSC-cardiomyocyte model is yet prominent in the literature reviewed. **Research applications:** mechanism of intercalated-disc/desmosome maintenance, arrhythmogenesis, inflammatory and oxidative-stress contributions, and preclinical testing of anti-inflammatory/antioxidant interventions.

Model resources: MGI (mouse), OMIA (cattle, natural disease).

Mechanistic Model / Interpretation



The **structural desmosomal lesion is the dominant, directly demonstrated driver** (human explants, mouse, cattle). The **inflammatory and oxidative branches are amplifiers**: the inflammatory arm is supported by human explant NF-κB data and the clinical response of arrhythmia to IL-1 blockade, while the oxidative arm is currently **inferred from rodent cardiomyocytes**. The **p63 branch** cleanly explains the non-cardiac ectodermal phenotype, unifying the "cardiocutaneous" syndrome under a single gene.

Evidence Base

PMID	Study type	Contribution
32666529	Human case series (GeneMatcher)	Gene–disease validation; 7 children/5 families; natural history & poor prognosis
28069640	Human cohort	First human description; founder nonsense allele; lethal infantile DCM
25691752	Mouse + in vitro	Core desmosomal mechanism; iASPP–desmoplakin–desmin; ARVC & sudden death
41928611	Human explant + therapy	Desmin-loss central mechanism; NF-κB in non-cardiomyocytes; IL-1 blockade
19016676	Cattle (natural model)	PPP1R13L frameshift causes bovine cardiomyopathy + woolly haircoat
21930934	In vitro	iASPP controls epithelial stratification via p63 (ectodermal features)
26646590	In vitro	iASPP is a common inhibitor of RelA/p65 NF-κB and p53
33332723	Rodent cardiomyocytes	iASPP cardioprotection via Keap1–Nrf2; ↓ ROS/apoptosis
37698259	Human family	Null allele p.Arg330ProfsTer76; intrafamilial variability; early deaths
35933355	Human case	p.Gln194Ter nonsense → ACM; recommends PPP1R13L on ACM panels
39579152	Review/case	Core triad; ~15 patients; extended phenotype (anorectal anomaly)
34970070	Review	CMR-LGE as imaging gold standard for ACM/DCM
42708185	ClinGen curation	PPP1R13L among curated DCM genes (2024–2025 reassessment)
30453078	Human panel screen	Monoallelic missense = VUS; underscores biallelic-LoF mechanism; TP63 link
42098522	Forensic	Postmortem genetic testing detects PPP1R13L in sudden death
18275817	Structural	iASPP ankyrin-repeat/SH3 architecture; differential p53/p63/p73 binding

PMID	Study type	Contribution
35924320	Human case	Expands phenotype; inflammation on histology; variably expressed ectodermal features

Limitations and Knowledge Gaps

1. **Ultra-rarity.** Conclusions rest on ~15–20 patients and a handful of families; no registry, prevalence, incidence, penetrance, or carrier-frequency estimates exist.
2. **Oxidative-stress arm is extrapolated.** The Keap1–Nrf2 cardioprotective mechanism is shown in rodent cardiomyocytes, not directly in human PPP1R13L hearts.
3. **Genotype–phenotype correlation is unresolved.** Why ectodermal features vary so widely, and what modifies cardiac severity, is unknown.
4. **No validated QoL/natural-history instruments** and limited long-term post-transplant outcome data.
5. **Therapeutics.** IL-1 blockade reduces arrhythmia but not pump failure; no gene/cell/RNA therapy exists. No controlled trials are feasible at current patient numbers.
6. **Human cellular models** (e.g., patient iPSC-cardiomyocytes) are not yet prominent, limiting mechanistic dissection in a human genetic background.

Proposed Follow-up Experiments / Actions

1. **Establish an international PPP1R13L registry** to aggregate genotype, phenotype, natural history, and transplant/mortality outcomes across centers.
2. **Patient-derived iPSC-cardiomyocyte models** to test, in human background, the relative contributions of the desmosomal, NF-κB/IL-1, and Nrf2/Keap1 arms and to screen candidate drugs.
3. **Prospective evaluation of IL-1 blockade** (and other anti-inflammatory/antioxidant agents, e.g., Nrf2 activators) with standardized arrhythmia and contractile endpoints.
4. **Deep phenotyping of ectodermal/ocular/orofacial features** with p63-pathway biomarkers to understand variable expressivity.

5. **Carrier-frequency and founder-allele mapping** in consanguineous/founder populations (e.g., Arab Christian communities) to inform screening.
6. **Preclinical gene-replacement or base-editing proof-of-concept** in Ppp1r13l-mutant mice, leveraging the recessive LoF mechanism.
7. **Advocate inclusion of PPP1R13L on all pediatric DCM/ACM gene panels** and promote postmortem genetic testing in unexplained pediatric sudden death.

Evidence source legend: Human clinical (case reports/series, explants, forensic), Model organism (mouse, cattle), In vitro (cardiomyocyte, epithelial, cancer cell), Computational/structural. This report synthesizes 8 confirmed findings and 27 reviewed papers from a 5-iteration autonomous investigation.
